

Exp No: 8
Date: 22.04.25

LIBRARY MANAGEMENT SYSTEM

AIM:

Write a program to implement a library management system

PROGRAM:

LIBRARYMANAGEMENT.JSP

```
<html>
<head>
  <title>Library Management - Add Book</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background-color: #f3f3f3;
      margin: 20px;
      padding: 20px;
    }

    h2 {
      color: #333;
    }

    form {
      background-color: #fff;
      padding: 20px;
      border-radius: 8px;
      box-shadow: 0 2px 5px rgba(0,0,0,0.1);
      width: 400px;
    }

    input[type="text"],
    input[type="number"],
    select {
      width: 100%;
      padding: 8px;
      margin: 6px 0 12px 0;
      border: 1px solid #ccc;
      border-radius: 4px;
    }

    input[type="submit"] {
      background-color: #4CAF50;
      color: white;
      border: none;
      padding: 10px 16px;
      cursor: pointer;
    }
  </style>
</head>
<body>
  <h2>Library Management - Add Book</h2>
  <form>
    <input type="text" value="Book Name" />
    <input type="text" value="Author Name" />
    <input type="text" value="ISBN" />
    <input type="text" value="Category" />
    <input type="text" value="Quantity" />
    <input type="submit" value="Add Book" />
  </form>
</body>
</html>
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    border-radius: 4px;
}

input[type="submit"]:hover {
    background-color: #45a049;
}

p {
    font-size: 16px;
}

a {
    color: #007BFF;
    text-decoration: none;
}

a:hover {
    text-decoration: underline;
}
</style>

<script>

function validateForm() {
    const bookId = document.getElementById("bookId").value.trim();
    const title = document.getElementById("title").value.trim();
    const author = document.getElementById("author").value.trim();
    const year = document.getElementById("year").value.trim();
    const category = document.getElementById("category").value;
    const copies = document.getElementById("copies").value.trim();
    const currentYear = new Date().getFullYear();

    if (!/^[a-zA-Z0-9]+$/.test(bookId)) {
        alert("Book ID must be alphanumeric.");
        return false;
    }
    if (!/^[A-Za-z ]+$/.test(title)) {
        alert("Title must contain only letters.");
        return false;
    }
    if (!/^[A-Za-z ]+$/.test(author)) {
        alert("Author name must contain only letters.");
        return false;
    }
    if (!/^\d{4}$/.test(year) || parseInt(year) > currentYear) {
        alert("Year must be a valid 4-digit year.");
        return false;
    }
}

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        if (category === "") {
            alert("Please select a category.");
            return false;
        }
        if (isNaN(copies) || parseInt(copies) <= 0) {
            alert("Copies must be a positive number.");
            return false;
        }

        return true;
    }
</script>
</head>
<body>
    <h2>Add Book to Library</h2>
    <form method="post" onsubmit="return validateForm()">
        Book ID: <input type="text" name="bookId" id="bookId"/><br/>
        Title: <input type="text" name="title" id="title"/><br/>
        Author: <input type="text" name="author" id="author"/><br/>
        Year: <input type="text" name="year" id="year"/><br/>
        Category:
        <select name="category" id="category">
            <option value="">--Select--</option>
            <option value="Science">Science</option>
            <option value="Fiction">Fiction</option>
            <option value="History">History</option>
        </select><br/>
        Copies: <input type="number" name="copies"
id="copies"/><br/><br/>
        <input type="submit" value="Add Book"/>

        <p>
            <label>Book Details:-</label>
            <a href="./display_books.jsp">Click Here</a>
        </p>
    </form>

    <%

        // JSP Database logic

        // TODO:Change password before run the code
        // TODO: CREATE THE TABLE IN MYSQL
        /**
CREATE TABLE books (
    book_id VARCHAR(20) PRIMARY KEY,
    title VARCHAR(100),
    author VARCHAR(100),
    year INT,

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        category VARCHAR(50),
        copies INT
    );
    **/

    String bookId = request.getParameter("bookId");
    if (bookId != null) {
        String title = request.getParameter("title");
        String author = request.getParameter("author");
        String year = request.getParameter("year");
        String isbn = request.getParameter("isbn");
        String category = request.getParameter("category");
        String copies = request.getParameter("copies");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection conn = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/library_management_akshayaas", "root",
                "");
            PreparedStatement ps = conn.prepareStatement(
                "INSERT INTO books (book_id, title, author, year,
category, copies) VALUES (?, ?, ?, ?, ?, ?)");
            ps.setString(1, bookId);
            ps.setString(2, title);
            ps.setString(3, author);
            ps.setInt(4, Integer.parseInt(year));
            ps.setString(5, category);
            ps.setInt(6, Integer.parseInt(copies));

            int result = ps.executeUpdate();
            if (result > 0) {
                %>
                <script>alert("Book Added Successfully!!")</script>
                <%
            } else {
                %>
                <script>alert("Failed to add book!!")</script>
                <%
                out.println("<p style='color:red;'>Failed to add
book.</p>");
            }

            ps.close();
            conn.close();
        } catch (Exception e) {
            out.println("<p style='color:red;'>Error: " + e.getMessage()
+ "</p>");
        }
    }
}

```

```
    }  
    %>  
</body>  
</html>
```

DISPLAY_BOOKS.JSP

```
<html>  
  
<head>  
  
  <title>Library Books List</title>  
  
  <style>  
  
    table {  
  
      border-collapse: collapse;  
  
      width: 90%;  
  
      margin: 20px auto;  
  
    }  
  
    th, td {  
  
      border: 1px solid #333;  
  
      padding: 10px;  
  
      text-align: center;  
  
    }  
  
    th {  
  
      background-color: #f2f2f2;  
  
    }  
  
    h2 {  
  
      text-align: center;  
  
    }  
  
  </style>
```

```
</head>

<body>

  <h2>All Books in Library</h2>

  <%

    try {

      Class.forName("com.mysql.cj.jdbc.Driver");

      Connection conn = DriverManager.getConnection(

        "jdbc:mysql://localhost:3306/library_management_akshayaas",

        "root", "");

      Statement stmt = conn.createStatement();

      ResultSet rs = stmt.executeQuery("SELECT * FROM books");

      out.println("<table>");

      out.println("<tr><th>Book  
ID</th><th>Title</th><th>Author</th><th>Year</th><th>Category</th><th>  
Copies</th></tr>");

      while (rs.next()) {

        out.println("<tr>");

        out.println("<td>" + rs.getString("book_id") + "</td>");

        out.println("<td>" + rs.getString("title") + "</td>");

        out.println("<td>" + rs.getString("author") + "</td>");

        out.println("<td>" + rs.getInt("year") + "</td>");

        out.println("<td>" + rs.getString("category") + "</td>");

        out.println("<td>" + rs.getInt("copies") + "</td>");
```

```

        out.println("</tr>");
    }

    out.println("</table>");

    rs.close();

    stmt.close();

    conn.close();

} catch (Exception e) {

    out.println("<p style='color:red;text-align:center;'>Error: "
+ e.getMessage() + "</p>");

}

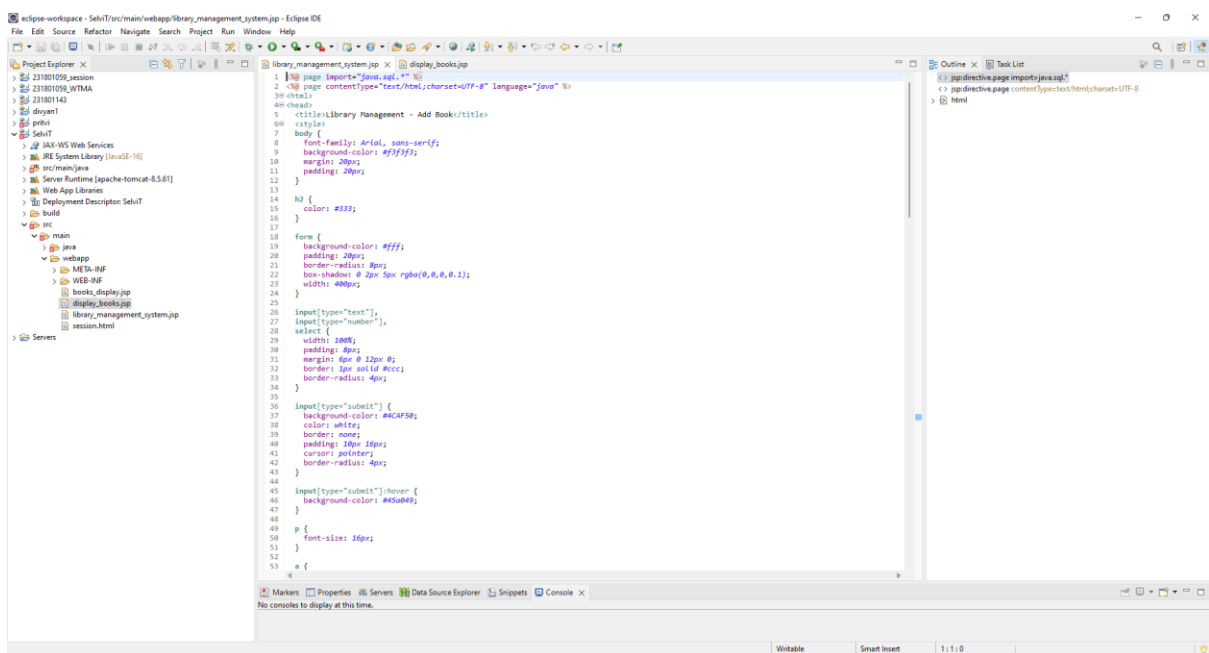
%>

</body>

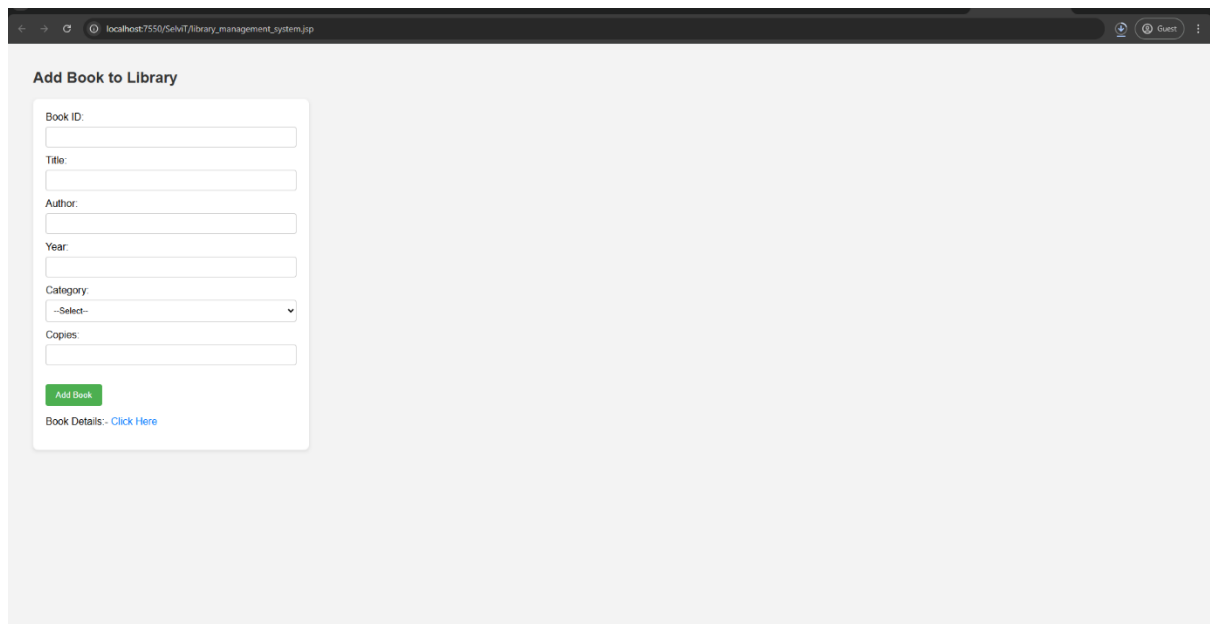
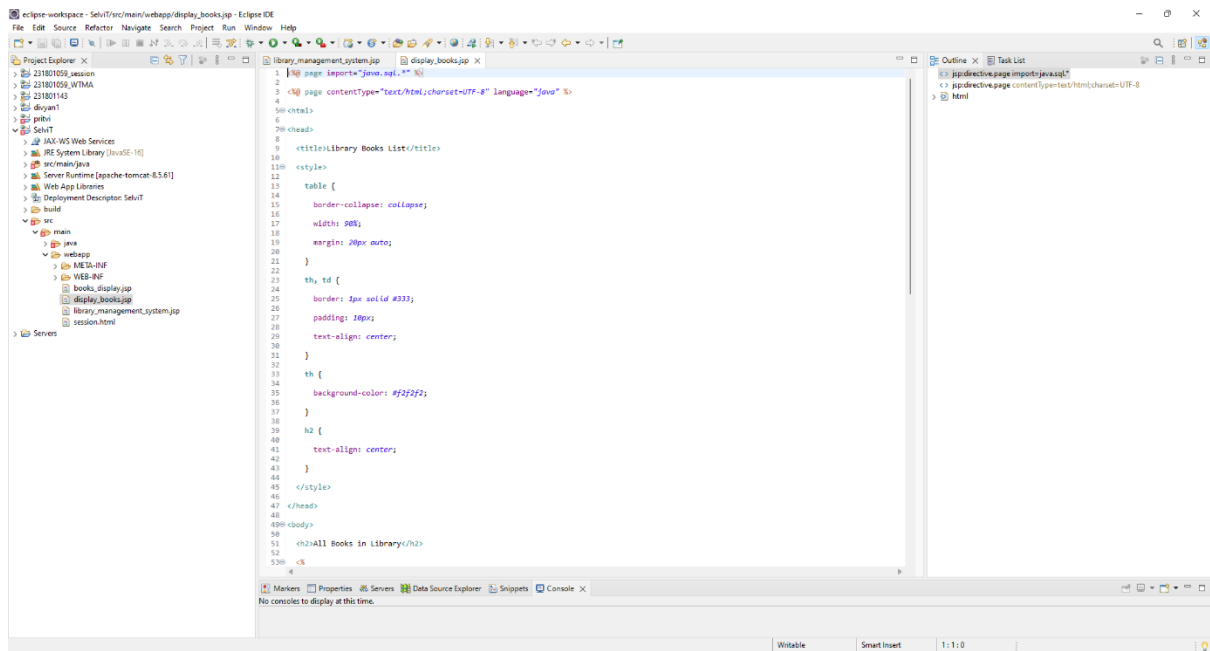
</html>

```

OUTPUT:



231801161



localhost:7550/SelviT/library_management_system.jsp

Guest

Add Book to Library

Book ID:
23178

Title:
Life

Author:
Jake

Year:
2025

Category:
Fiction

Copies:
1000

[Add Book](#)

Book Details - [Click Here](#)

RESULT :

Thus the experiment to implement the library management system has been successfully implemented and verified

231801161